

IMECE2026-192449

**MULTI-CYCLE POLYMER INFILTRATION AND PYROLYSIS PROCESSING OF
CARBON FIBER REINFORCED CERAMIC MATRIX COMPOSITES FOR HIGH-
TEMPERATURE AEROSPACE APPLICATIONS**

Bailey Walke
KANSAS STATE UNIVERSITY
Manhattan, KS

Dr. Gurpreet Singh
KANSAS STATE UNIVERSITY
Manhattan, KS

ABSTRACT

Ceramic matrix composites (CMCs) produced via polymer infiltration and pyrolysis (PIP) offer a scalable pathway for high-temperature aerospace applications; however, traditional multi-cycle PIP processing is time-intensive due to repeated high-temperature pyrolysis steps. This work investigates a modified multi-infiltration single-pyrolysis approach to reduce processing time while maintaining composite performance.

Carbon fiber bundles were infiltrated with polysilazane and boron-modified polysilazane precursors to form SiCN and SiBCN ceramic matrices. Two processing routes were compared: a traditional three-cycle PIP method with repeated pyrolysis and a modified method consisting of multiple infiltration and crosslinking cycles followed by a single final pyrolysis.

Processing time analysis demonstrates a significant reduction in total fabrication time using the modified method. Mass and ceramic yield measurements confirm effective ceramic matrix formation in both approaches. Oxidation testing at 800°C shows comparable performance between methods when normalized to ceramic content, with differences attributed to ceramic loading. Scanning electron microscopy reveals reduced microcracking and more uniform coating morphology in samples processed with the modified method. Tensile testing indicates that mechanical behavior is preserved, with differences governed primarily by ceramic-to-fiber ratio.

These results demonstrate that the modified PIP approach improves manufacturing efficiency without sacrificing composite performance, supporting its potential for scalable CMC production.

Keywords: Ceramic matrix composites, polymer infiltration and pyrolysis, polymer-derived ceramics, SiCN, SiBCN, oxidation resistance, manufacturing scalability

1. INTRODUCTION

CMCs are advanced structural materials designed for high-temperature and extreme environment applications where conventional metallic alloys and polymer matrix composites are unable to maintain performance. By combining high-strength ceramic or carbon fiber reinforcement with a ceramic matrix, CMCs offer a unique balance of low density, high-temperature strength retention, oxidation resistance, and improved damage tolerance compared to monolithic ceramics [1–3]. These properties have led to increasing adoption of CMCs in aerospace propulsion systems, gas turbines, hypersonic vehicles, and thermal protection systems, where elevated operating temperatures can directly improve system efficiency and performance [1,4,5].

Among the various fabrication methods for CMCs, including chemical vapor infiltration (CVI) and melt infiltration (MI), polymer infiltration and pyrolysis (PIP) is particularly attractive due to its relatively low processing temperature, ability to fabricate complex geometries, and compatibility with a wide range of preceramic polymer precursors [5,9]. In the PIP process, a liquid polymer precursor infiltrates a porous fiber preform, is crosslinked into a solid network, and is subsequently pyrolyzed at elevated temperature to form a ceramic matrix. However, polymer-to-ceramic conversion is accompanied by significant shrinkage and porosity formation, resulting in low ceramic yield per cycle and requiring multiple infiltration and pyrolysis cycles to achieve sufficient densification [10,11].

While increasing the number of PIP cycles improves matrix density and mechanical properties, repeated high-temperature pyrolysis steps significantly increase processing time, energy consumption, and manufacturing cost. In addition, repeated thermal cycling can introduce matrix shrinkage and microcracking, which may negatively impact microstructural integrity and composite performance [10,11]. These limitations present a significant challenge to the scalability of PIP-based

manufacturing for large-scale or high-throughput production of CMC components.

Polymer-derived ceramics such as silicon carbonitride (SiCN) and silicon boron carbonitride (SiBCN) are of particular interest due to their thermal stability, oxidation resistance, and amorphous nanocomposite structure at elevated temperatures [6–8]. When combined with carbon fiber reinforcement, these materials can provide lightweight composites with improved environmental resistance. However, the relationship between processing approach, ceramic matrix formation, and resulting material performance must be understood to enable more efficient manufacturing strategies.

In this work, a modified polymer infiltration and pyrolysis approach is investigated to reduce processing time while maintaining composite performance. A traditional multi-cycle PIP method consisting of repeated infiltration, crosslinking, and pyrolysis steps is compared to a multi-infiltration single-pyrolysis method, in which multiple infiltration and crosslinking cycles are performed prior to a single final pyrolysis step. The two methods are evaluated through processing time analysis, ceramic yield measurements, oxidation testing, scanning electron microscopy, and tensile testing to assess differences in matrix formation, microstructure, environmental stability, and mechanical behavior.

The objective of this study is to determine whether reducing the number of pyrolysis cycles can significantly improve manufacturing efficiency without compromising material performance. Demonstrating this capability would support the development of more scalable polymer-derived CMC manufacturing processes for high-temperature aerospace and energy applications.

2. MATERIALS AND METHODS

2.1 Materials

Carbon fiber cloth (6K, 2×2 twill weave, 0.017 in thickness, 10.9 oz/yd², Fibre Glast Developments Corporation) was used as the reinforcement material. Polysilazane (Ceraset PSZ 20, Clariant Corporation) served as the preceramic polymer precursor for SiCN fabrication. Isopropanol (≥99.9%, Fisher Scientific) was used as a solvent to reduce precursor viscosity for infiltration. For SiBCN samples, trimethyl borate (TMB, ≥98%, Sigma-Aldrich) was added to the polysilazane precursor to introduce boron into the ceramic network. All precursor preparation and infiltration processes were performed in an oxygen-free glovebox environment to prevent premature oxidation and improve ceramic yield.

2.2 Preceramic Precursor Preparation

For SiCN fabrication, polysilazane was diluted with isopropanol at a 4:3 weight ratio to achieve a viscosity suitable for infiltration into carbon fiber bundles. The solution was stirred at 300 rpm overnight to ensure uniform mixing.

For SiBCN fabrication, trimethyl borate was added to the polysilazane precursor at approximately a 1:1 weight ratio

following established procedures [12]. The mixture was stirred at 300 rpm overnight to promote homogeneous boron incorporation into the polymer network. The polymer precursor preparation process is visualized in Figure 1.

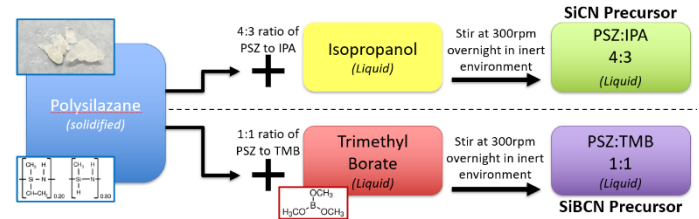


FIGURE 1: Schematic representation of preceramic precursor preparation for SiCN and SiBCN systems.

2.3 Composite Fabrication

Carbon fiber bundles were cut into approximately 5 cm long samples with a mass of ~10 mg. Preceramic polymer precursor was infiltrated into each bundle using a micropipette at a ratio of approximately 1 μ L per 1 mg of fiber to ensure consistent polymer loading. After infiltration, samples were crosslinked at 180 °C for 1 hour in an inert environment to stabilize the polymer prior to pyrolysis.

Following crosslinking, pyrolysis was performed in a nitrogen atmosphere at 800°C for 30 minutes to convert the polymer into a ceramic matrix. The complete thermal cycle required approximately 8 hours.

Two processing approaches were investigated, differing in the number and sequence of infiltration and pyrolysis steps.

2.3.1 Method 1: Traditional Multi-Cycle PIP

In the traditional polymer infiltration and pyrolysis (PIP) process, infiltration, crosslinking, and pyrolysis were performed sequentially for each cycle. After each pyrolysis step, samples were re-infiltrated with precursor, crosslinked, and pyrolyzed again (Figure 2). This multi-cycle process was repeated for three cycles to increase ceramic matrix content and reduce porosity within the composite.

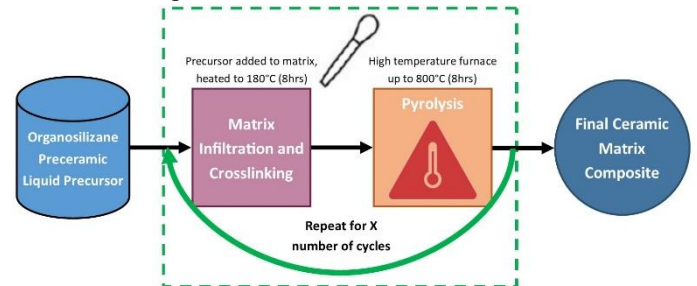


FIGURE 2: Traditional multi-cycle PIP process, where infiltration, crosslinking and pyrolysis are repeated to increase ceramic matrix density.

2.3.2 Method 2: Multi-Infiltration Single-Pyrolysis

In the modified processing approach, multiple infiltration and crosslinking cycles were performed prior to a single final pyrolysis step (Figure 3). After each infiltration, samples were

crosslinked at 180 °C for 1 hour, and the infiltration–crosslinking sequence was repeated to increase polymer loading within the fiber bundles. Following completion of the desired number of infiltration cycles, a single pyrolysis step at 800°C was performed to convert the accumulated polymer into a ceramic matrix.

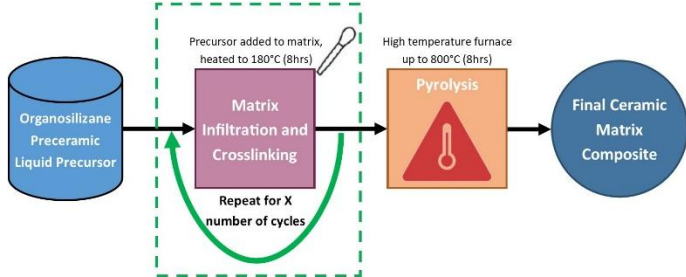


FIGURE 3: Modified PIP process consisting of multiple infiltration and crosslinking cycles followed by a single pyrolysis step.

2.4 Characterization Methods

Scanning electron microscopy (SEM) was used to evaluate ceramic coating formation, matrix distribution, and microstructural features of the composites. Oxidation resistance was assessed by exposing samples to 800°C in air, with mass retention used to quantify environmental stability. Room-temperature tensile testing of mini-composite bundles was conducted to evaluate mechanical behavior relative to uncoated carbon fiber and between processing methods. Additional compositional characterization using X-ray-based techniques was performed where necessary to confirm ceramic formation and elemental composition.

3. RESULTS AND DISCUSSION

3.1 Processing Time Reduction

The modified multi-infiltration single-pyrolysis processing method resulted in a significant reduction in total fabrication time compared to the traditional multi-cycle polymer infiltration and pyrolysis (PIP) approach. The traditional method, consisting of three full infiltration–crosslinking–pyrolysis cycles, required approximately 30 hours of total processing time. In contrast, the modified method reduced total processing time to approximately 14 hours, corresponding to a reduction of approximately 53%.

This reduction is primarily attributed to the elimination of repeated high-temperature pyrolysis cycles, which represent the most time- and energy-intensive portion of the PIP process (Figure 8). While infiltration and crosslinking steps remain unchanged, these processes occur at significantly lower temperatures and contribute minimally to overall processing time compared to furnace-based pyrolysis.

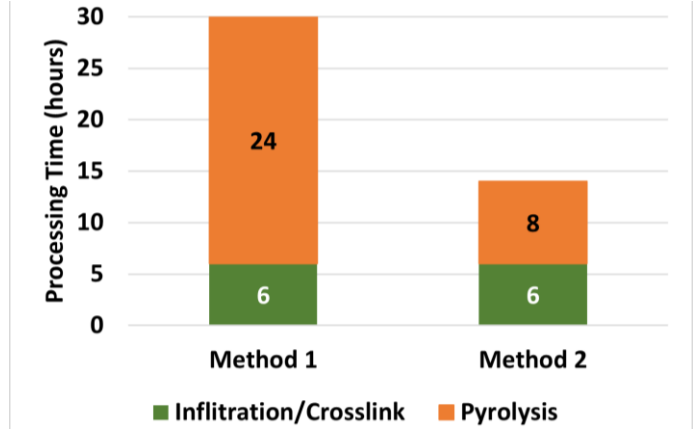


FIGURE 4: Comparison of total processing time for traditional multi-cycle PIP (Method 1) and modified multi-infiltration single-pyrolysis (Method 2), highlighting the reduction in high-temperature processing steps.

Despite the reduced processing time, the modified method maintains comparable performance when accounting for differences in ceramic loading. The additional pyrolysis cycles in the traditional method provide diminishing returns in terms of mechanical and oxidation performance, indicating that repeated ceramic conversion is not the primary driver of property improvement.

From a manufacturing perspective, the reduction in processing time has significant implications for scalability. Shorter processing cycles increase production throughput, reduce energy consumption, and minimize reliance on high-temperature furnace operations, which are a key bottleneck in large-scale CMC manufacturing.

3.2 Mass and Ceramic Yield Analysis

Mass measurements were used to evaluate polymer infiltration effectiveness, ceramic accumulation, and final composite composition for both processing methods (Table 1).

Ceramic yield and ceramic ratio were calculated as:

$$\text{Ceramic Yield (\%)} = \frac{m_{\text{ceramic}}}{m_{\text{polymer}}} \times 100 \quad (1)$$

$$\text{Ceramic Ratio (\%)} = \frac{m_{\text{ceramic}}}{m_{\text{final}}} \times 100 \quad (2)$$

Across both methods, SiCN-CF samples exhibited significantly greater polymer uptake than SiBCN-CF samples, resulting in higher final mass and increased ceramic content. In the modified process (Method 2), total preceramic polymer addition ranged from approximately 16–22 mg for SiCN-CF compared to 7–9 mg for SiBCN-CF, corresponding to substantially higher pre-pyrolysis polymer ratios in the SiCN system. Following pyrolysis, this difference translated directly to final composite composition, with SiCN-CF samples achieving approximately 3–4× greater ceramic accumulation than SiBCN-CF samples.

Although ceramic yield values were comparable between systems, the results demonstrate that conversion efficiency alone does not govern final ceramic content. Instead, the dominant

factor controlling composite composition is the total amount of polymer successfully retained within the fiber preform prior to pyrolysis. This behavior was consistent across both processing methods and reflects differences in precursor infiltration behavior. The SiBCN precursor, with observed lower viscosity, exhibited reduced retention within the fiber bundles, leading to lower overall polymer uptake. In contrast, the higher viscosity SiCN precursor promoted retention and resulted in greater ceramic accumulation after pyrolysis.

Final composition analysis further highlights this distinction. SiCN-CF composites exhibited ceramic fractions of approximately 55–65%, indicating a transition toward ceramic-dominated behavior, while SiBCN-CF composites remained fiber-dominated with ceramic fractions below approximately 40%. These trends were consistent for both processing methods, indicating that polymer infiltration effectiveness, not the number of pyrolysis cycles, is the primary factor governing matrix formation and final composite structure.

TABLE 1: Comparison of polymer uptake, ceramic yield, and final composition for Method 1 (three-cycle PIP) and Method 2 (multi-infiltration single-pyrolysis) samples.

Method	Material	Preceramic Polymer Added (mg)	Ceramic Material Added (mg)	Ceramic Yield (%)	Ceramic Ratio (%)
M1 (3rd Cycle)	SiCN-CF	30.3	20.1	66.3	67.2
	SiBCN-CF	8.7	6.5	74.7	39.9
M2	SiCN-CF	19.8	14.3	73.8	59.1
	SiBCN-CF	7.9	3.3	41.4	24.8

Comparison between processing methods shows that similar relationships between polymer uptake and final ceramic content are maintained despite the reduced number of pyrolysis cycles in Method 2. While the traditional multi-cycle PIP process provides incremental increases in ceramic content, diminishing mass uptake in later cycles indicates that additional processing yields progressively smaller gains. The modified method achieves comparable levels of ceramic accumulation using a simplified processing sequence, demonstrating that repeated high-temperature pyrolysis is not required to achieve effective matrix formation.

3.3 Oxidation Performance

Oxidation testing was performed to evaluate the high-temperature stability of SiCN-CF and SiBCN-CF composites produced using both processing methods. All ceramic-coated samples remained structurally intact following exposure to 800°C in air, while the uncoated carbon fiber control sample completely disintegrated, confirming the formation of an effective protective ceramic barrier.

Mass retention and protective efficiency were calculated as:

$$\text{Mass Retention (\%)} = \frac{m_{\text{post-ox}}}{m_{\text{final}}} \times 100 \quad (3)$$

$$\text{Protective Efficiency (\%)} = \frac{m_{\text{post-ox}}}{m_{\text{ceramic}}} \times 100 \quad (4)$$

TABLE 2: Summary of oxidation performance for SiCN-CF and SiBCN-CF composites processed using Method 1 (multi-cycle PIP) and Method 2 (multi-infiltration single-pyrolysis).

Material	Method	Cycles	Ceramic Mass (mg)	Mass Retention (%)	Protective Efficiency (%)
SiCN-CF	M1	1	5.8	27.33	75.86
		2	11.4	48.11	89.47
		3	20.1	69.57	103.48
	M2	—	13.2	36.21	63.64
SiBCN-CF	M1	1	2.2	19.35	109.09
		2	4.5	29.37	93.33
		3	6.5	36.20	90.77
	M2	—	4.2	22.38	76.19

Mass retention results (Table 2) show a strong dependence on ceramic loading across both material systems and processing methods. Samples with higher ceramic content consistently exhibited greater oxidation resistance, indicating that increased matrix continuity reduces oxygen diffusion and limits degradation. SiCN-CF samples, which achieved significantly higher ceramic loading, demonstrated superior absolute oxidation resistance, with mass retention values reaching approximately 69.6% for the multi-cycle PIP method and 36.2% for the modified method. In comparison, SiBCN-CF samples exhibited lower mass retention, with values of approximately 36.2% and 22.4% for the two methods, respectively.

However, when normalized by ceramic content, an opposing trend is observed. SiBCN-CF samples exhibited higher mass retention efficiency, indicating more effective oxidation protection per unit of ceramic material. This suggests that while SiCN benefits from higher ceramic loading, SiBCN provides more efficient surface protection, likely due to differences in oxide scale formation and glassy phase behavior. Values exceeding 100% are attributed to measurement uncertainty and potential mass contributions from oxide formation during high-temperature exposure.

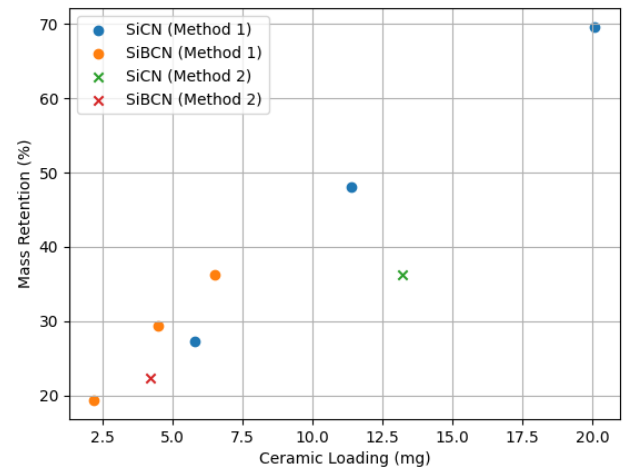


FIGURE 5: Relationship between ceramic loading and oxidation mass retention for SiCN-CF and SiBCN-CF composites, showing increased absolute performance with higher ceramic content and improved efficiency at lower ceramic loading.

Comparison between processing methods further demonstrates that oxidation performance is governed primarily by ceramic loading rather than the number of pyrolysis cycles. Despite reduced processing time, Method 2 samples follow the same relationship between ceramic content and oxidation resistance as Method 1 samples. This indicates that comparable oxidation performance can be achieved without repeated high-temperature processing, provided sufficient polymer infiltration is achieved prior to pyrolysis.

Overall, oxidation behavior is controlled by a trade-off between total ceramic content and material efficiency. SiCN-CF composites provide superior absolute oxidation resistance due to higher ceramic loading, while SiBCN-CF composites offer more efficient protection at lower ceramic content. These results reinforce that polymer infiltration effectiveness, not repeated pyrolysis cycles, is the dominant factor governing environmental performance.

3.4 Microstructural Analysis

SEM was used to compare ceramic coating formation, matrix distribution, and microstructural damage in pyrolyzed SiCN-CF (Figure 6) and SiBCN-CF (Figure 7) composites fabricated using both processing methods. This comparison focuses on the post-pyrolysis condition to provide a direct evaluation of the processing–microstructure relationship between the traditional multi-cycle PIP method and the modified multi-infiltration single-pyrolysis method.

For Method 1 samples, ceramic coating thickness and coverage increased with increasing PIP cycle number. Lower cycle samples exhibited thin, discontinuous coatings with regions of exposed carbon fiber, while higher cycle samples showed thicker and more continuous matrix formation. However, this increased matrix buildup was accompanied by non-uniform coating thickness, surface roughness, and interconnected microcracking, particularly in higher-cycle SiCN-CF samples. These defects are attributed to cumulative shrinkage stresses generated during repeated polymer-to-ceramic conversion and multiple high-temperature pyrolysis cycles.

In contrast, Method 2 samples exhibited smoother and more continuous ceramic coatings along individual fibers, with improved fiber definition and fewer large-scale crack networks. Because the modified route performs multiple infiltration and crosslinking steps prior to a single pyrolysis cycle, polymer conversion occurs once after greater precursor loading has been achieved. This reduces the number of shrinkage events and limits the accumulation of thermal processing damage. Localized matrix accumulation and microcracking were still observed, especially in SiCN-CF samples, but these defects were less severe and less interconnected than those observed in Method 1.

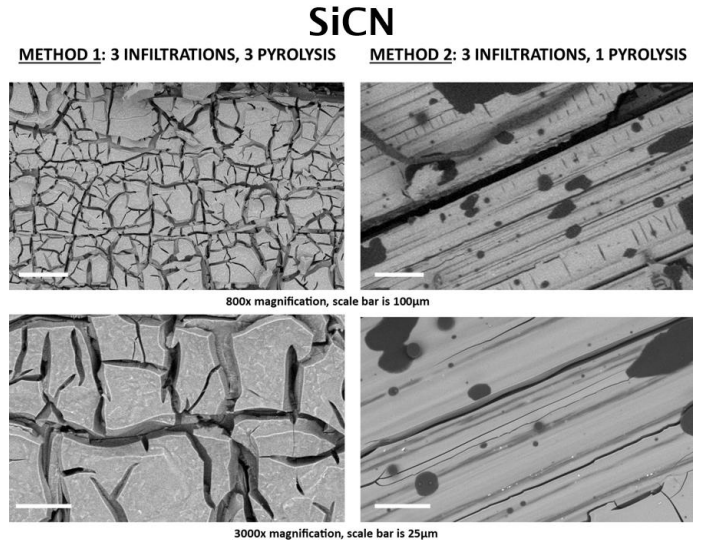


FIGURE 6: Pyrolyzed SEM comparison of SiCN-CF composites fabricated using Method 1 and Method 2, showing increased matrix buildup and crack formation in Method 1 compared to more continuous coating morphology in Method 2.

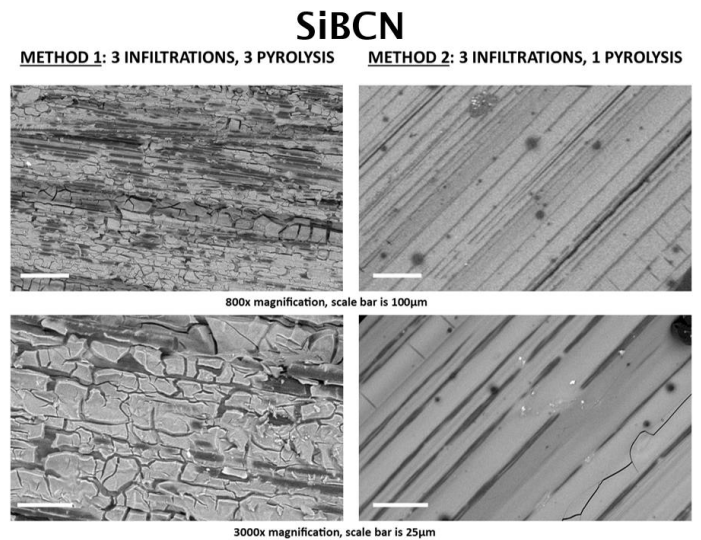


FIGURE 7: Pyrolyzed SEM comparison of SiBCN-CF composites fabricated using Method 1 and Method 2, showing improved fiber definition and more uniform coating distribution in Method 2.

Material-dependent differences were also observed. SiCN-CF samples generally exhibited greater ceramic buildup, consistent with the higher ceramic loading measured in the mass analysis. This increased ceramic content improved matrix formation but also increased the tendency for shrinkage-induced cracking. SiBCN-CF samples showed clearer fiber definition and more uniform coating distribution, suggesting improved wetting and precursor distribution, but with lower overall ceramic accumulation.

Comparing both samples similar images of high-cycle PIP processing methods (8-80 infiltrations), Method 2 shows similar dense matrix formation and complete fiber coverage [13].

Overall, SEM analysis shows that both processing methods produce ceramic coatings on carbon fiber bundles, but the modified Method 2 route achieves comparable or improved coating continuity with reduced microstructural damage. This supports the conclusion that delaying ceramic conversion until after multiple infiltration steps can reduce shrinkage-induced cracking while maintaining effective ceramic matrix formation.

3.5 Mechanical Properties

Tensile testing was performed to evaluate the mechanical response of SiCN-CF and SiBCN-CF composites fabricated using the modified multi-infiltration single-pyrolysis process. Mechanical property results of CMC samples and a neat carbon fiber control are reported in Table 3. The incorporation of a ceramic matrix significantly altered the mechanical behavior compared to neat carbon fiber, increasing stiffness while reducing ductility and toughness.

Neat carbon fiber exhibited the highest ultimate tensile strength (116 MPa) and failure strain (0.015), indicating strongly fiber-dominated behavior with gradual failure. In contrast, both SiCN-CF and SiBCN-CF composites showed increased elastic modulus (17.1 GPa and 18.6 GPa, respectively) but reduced strain to failure due to the presence of a brittle ceramic matrix.

TABLE 3: Mechanical properties of neat carbon fiber, SiCN-CF, and SiBCN-CF composites processed using Method 2.

Material	Elastic Modulus (GPa)	UTS (MPa)	Failure Strain	Toughness (MJ/m ³)
Neat CF	14.133	116.023	0.015	4.903
SiCN-CF	17.063	32.333	0.002	0.051
SiBCN-CF	18.558	75.971	0.005	0.267

The stress–strain behavior for SiCN and SiBCN composites is shown in Figure 8. For the SiCN-CF system, Method 2 samples exhibit a steeper initial slope but fail at significantly lower strain compared to single-cycle samples, indicating increased stiffness but more brittle behavior. This response is attributed to the higher ceramic content in the SiCN composites, which promotes early crack initiation and limits fiber-dominated load transfer.

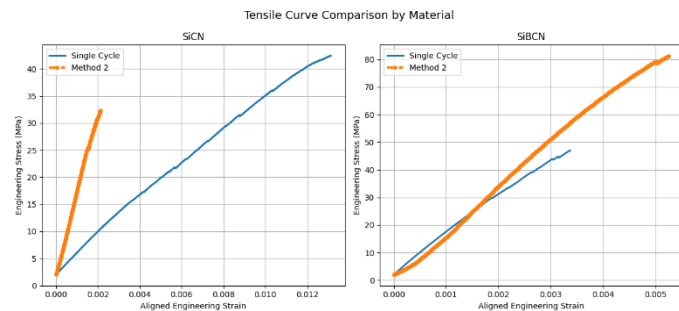


FIGURE 8: Stress–strain comparison of single-cycle and Method 2 composites for (a) SiCN-CF and (b) SiBCN-CF, illustrating the transition from fiber-dominated to ceramic-influenced mechanical behavior.

In contrast, SiBCN-CF composites demonstrate improved strength and more progressive failure behavior with Method 2

processing. The stress–strain curves show higher load capacity and extended deformation compared to single-cycle samples, indicating effective load transfer and continued fiber participation during fracture. This behavior reflects a more balanced fiber-to-ceramic ratio, where the ceramic matrix enhances stiffness without fully suppressing fiber-dominated mechanisms.

These results demonstrate that mechanical behavior is governed primarily by the relative phase balance between carbon fiber and ceramic matrix. Higher ceramic content increases stiffness but promotes brittle failure, while lower ceramic content preserves fiber-dominated mechanisms and improves strength and toughness. This trend is consistent with the mass analysis results, where SiCN-CF composites exhibited higher ceramic fractions and SiBCN-CF composites remained more fiber-dominated.

Overall, the modified processing method produces composites with mechanical behavior consistent with their composition, without introducing additional degradation due to reduced processing steps. This confirms that the multi-infiltration single-pyrolysis approach maintains structural integrity without introducing performance degradation despite reduced processing.

4. CONCLUSION

The modified multi-infiltration single-pyrolysis PIP method reduced processing time by approximately 53% while maintaining comparable oxidation resistance and mechanical performance. Results demonstrate that ceramic loading governs absolute performance, while processing strategy controls microstructural integrity and efficiency. Method 2 reduces shrinkage-induced damage and improves coating uniformity by minimizing repeated pyrolysis cycles. These findings support a shift toward optimizing polymer infiltration efficiency rather than maximizing ceramic content for scalable CMC manufacturing.

ACKNOWLEDGEMENTS

I would like to acknowledge Dr. Gurpreet Singh for support on this research and Arijit Roy for assistance with data collection.

This research is based on work supported by the National Science Foundation, under grant no. 1743701 to G. Singh.

REFERENCES

- [1] Krenkel, Walter, ed. Ceramic matrix composites: fiber reinforced ceramics and their applications. John Wiley & Sons, 2008.
- [2] Naslain, Roger. "Design, preparation and properties of non-oxide CMCs for application in engines and nuclear reactors: an overview." Composites Science and Technology 64.2 (2004): 155-170.
- [3] Chawla, Krishan K. Ceramic matrix composites. Springer Science & Business Media, 2013
- [4] Dever, Joyce A., Michael V. Nathal, and James A. DiCarlo. "Research on high-temperature aerospace materials at NASA

Glenn Research Center." *Journal of Aerospace Engineering* 26.2 (2013): 500-514.

[5] Shrivastava, Shriya, et al. "Ceramic matrix composites: classifications, manufacturing, properties, and applications." *Ceramics* 7.2 (2024): 652-679.

[6] Colombo, Paolo, et al. "Polymer-derived ceramics: 40 years of research and innovation in advanced ceramics." *Journal of the American Ceramic Society* 93.7 (2010): 1805-1837.

[7] Ren, Zhongkan, Shakir Bin Mujib, and Gurpreet Singh. "High-temperature properties and applications of Si-based polymer-derived ceramics: a review." *Materials* 14.3 (2021): 614.

[8] Riedel, Ralf, et al. "Silicon-based polymer-derived ceramics: synthesis properties and applications-a review dedicated to Prof. Dr. Fritz Aldinger on the occasion of his 65th birthday." *Journal of the Ceramic Society of Japan* 114.1330 (2006): 425-444.

[9] Santhosh, Unni, et al. "A polymer infiltration and pyrolysis (PIP) process model for ceramic matrix composites (CMCs)." *Journal of the American Ceramic Society* 104.12 (2021): 6108-6130.

[10] Venkatachalam, Vinothini, Burkard Esser, and Jon Binner. "Processing and performance of ultra high temperature ceramic matrix composite (UHTCMCs) using radio frequency assisted chemical vapour infiltration (RF-CVI)." *Composites Part A: Applied Science and Manufacturing* 185 (2024): 108358.

[11] Corman, G. S., and K. L. Luthra. "Silicon melt infiltrated ceramic composites (HiPerComp™)." *Handbook of ceramic composites*. Boston, MA: Springer US, 2005. 99-115.

[12] Singh, Gurpreet, and Romil Bhandavat. "Boron-modified silazanes for synthesis of SiBNC ceramics." U.S. Patent No. 9,453,111. 27 Sep. 2016.

[13] Zoli, Luca, et al. "Additive manufacturing of ceramics enabled by flash pyrolysis of polymer precursors with nanoscale layers." *Journal of the American Ceramic Society* 99.1 (2016): 57-63.